## BOUNDED STUDY INDEX | XR-ACCEL-01 | REFERENCE: VECTOR CPU EXTENSION

**CLAIM UNDER REVIEW** The shared-memory SoC block improves XR efficiency over a vector CPU extension and a loosely coupled accelerator within 3 W, preserving QoS and deployability.

### Intent

Improve mobile XR efficiency without violating power, QoS, memory, or deployment constraints.

### Task

Compare vector CPU extension (reference), loosely coupled accelerator, and shared-memory SoC block.

### Design space

Legal accelerator, memory, and vector choices; software redesign and full SoC changes are out.

### Representation

XRBench traces, architecture parameters, compiler assumptions, power model, and uncertainty record.

### Environment

Simulator or cost model plus workload harness; the contract defines actions and observations.

### Method role

Generate candidates, predict cost, optimize, challenge assumptions, and explain mechanisms.

### Feedback budget

Many cheap proxies, fewer simulations, and two high-fidelity checks before claim review.

### Evidence

Links to runs, three-way comparisons, sensitivities, provenance, and limits in the evidence record.

### Failed / rejected

Links to power violations, QoS misses, bandwidth overruns, proxy mismatches, and discarded options.

### Rejection checks + authority

Power, QoS, proxy, and cross-tool checks may block or weaken the claim; the architect sets disposition.

### Evidence-supported claim boundary

The evidence supports a higher-fidelity study only; no RTL, integration, or product commitment.

### Decision and owner

The named architect owns scope changes, waivers, claim strength, escalation, and commitment.

The card indexes the study, claim, artifacts, evidence, and decision. It does not replace the records.